Mining GOLD Samples for Conditional GANs

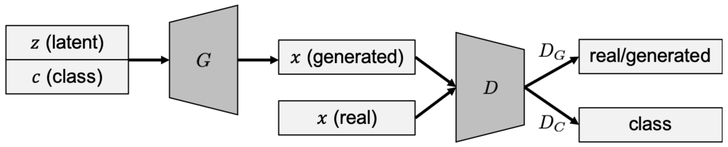
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We propose a quality measure for the real/generated samples of conditional GANs (cGANs), and utilize it to improve training, inference and labeling cost of cGANs.

Full paper | Code

Preliminary: Conditional GANs

Figure 1: Illustration of the architecture of ACGAN [1]



We mainly focus on the ACGAN[a] [1] which consists of

- the generator $G : (z, c) \mapsto x$ maps a pair of latent z and class c to a sample x
- the discriminator $D : x \mapsto (\{\text{real/generated}\}, c)$ consisting of
  - the GAN branch $D_G : x \mapsto \{\text{real/generated}\}$ which learns the marginal $p(x)$
  - the classifier branch $D_C : x \mapsto c$ which learns the conditional $p(c|x)$

ACGAN is trained by both the GAN loss $\mathcal{L}_{\text{GAN}}$ and the classifier loss $\mathcal{L}_{\text{AC}}$:[b,c]

$$\mathcal{L}_{\text{GAN}} = \mathbb{E}_{p_{\text{data}}(x,c)}[-\log D_G(x)] + \mathbb{E}_{p_g(z,c)}[\log D_G(G(z,c))]$$
$$\mathcal{L}_{\text{AC}} = \mathbb{E}_{p_{\text{data}}(x,c)}[-\log D_C(c|x)] + \lambda_c \mathbb{E}_{p_g(z,c)}[-\log D_C(c|G(z,c))]$$

[a]We choose ACGAN since it is more suitable for the limited label setting (semi-supervised learning). However, we remark that our main idea is generally applicable to the other types of cGANs, *e.g.*, projection discriminator [2] provides the other estimator form of the gap of log-densities.
[b]The generator and discriminator minimizes $-\mathcal{L}_{\text{GAN}} + \mathcal{L}_{\text{AC}}$ and $\mathcal{L}_{\text{GAN}} + \mathcal{L}_{\text{AC}}$, respectively.
[c]The original work [1] simply sets $\lambda_c = 1$, but we observe that using small λ_c works better.

Reference:
[1] Odena et al. Conditional Image Synthesis With Auxiliary Classifier GANs. ICML 2017.
[2] Miyato and Koyama. cGANs with Projection Discriminator. ICLR 2018.
[3] Lesort et al. Training Discriminative Models to Evaluate Generative Ones. arXiv 2018.
[4] Ravuri and Vinyals. Classification Accuracy Score for Conditional Generative Models. NeurIPS 2019.

Gap of log-densities (GOLD) estimator

We measure the suboptimality of the samples of cGANs by the discrepancy between the true distribution $p_{\text{data}}(x, c)$ and the model distribution $p_g(x, c)$

In particular, we consider the gap of log-densities (GOLD):

$$\log p_{\text{data}}(x,c) - \log p_g(x,c) = \underbrace{\log \frac{p_{\text{data}}(x)}{p_g(x)}}_{\text{marginal}} + \underbrace{\log \frac{p_{\text{data}}(c|x)}{p_g(c|x)}}_{\text{conditional}}, \quad (1)$$

which can be rewritten as the marginal and conditional terms: they measure the generation quality and class accuracy of the generated samples, respectively

We approximate (1) using two branches of the discriminator: D_G and D_C:

- Assuming optimal D_G^*, we approximate the marginal term by $\log \frac{D_G(x)}{1-D_G(x)}$
- Assuming optimal D_C^*, we approximate the conditional term as follow:[a]
  - For real sample x, let $p_{\text{data}}(c_x|x) = 1$ and $p_g(c_x|x) \approx D_C(c_x|x)$
  - For generated sample x, let $p_g(c_x|x) = 1$ and $p_{\text{data}}(c_x|x) \approx D_C(c_x|x)$

To sum up, the GOLD estimator is given by[b,c]

$$d(x, c_x) := \begin{cases} \log \frac{D_G(x)}{1-D_G(x)} + \log D_C(c_x|x) & \text{if } x \text{ is a generated sample} \\ \log \frac{D_G(x)}{1-D_G(x)} - \log D_C(c_x|x) & \text{if } x \text{ is a real sample} \end{cases} \quad (2)$$

[a]We denote c_x be the ground-truth label of a (real or generated) sample x.
[b]Note that the conditional term of (2) has an opposite sign for the generated and real samples. This matches the signs of marginal and conditional terms for both samples.
[c]We also propose a variant of the GOLD estimator, balancing marginal and conditional terms.

Applications of the GOLD estimator

Improving training of cGAN (example re-weighting):

- Guide the generator to encourage the under-estimated (high GOLD) regions, and suppress the over-estimated (low GOLD) regions
- We propose a re-weighted loss calibrated by the GOLD estimator:[a]

$$\mathcal{L}'_{\text{GAN}} = \mathbb{E}_{p_{\text{data}}(x,c)}[-\log D_G(x)] + \mathbb{E}_{p_g(z,c)}[d(G(z,c),c)^\beta \log D_G(G(z,c))]$$
$$\mathcal{L}'_{\text{AC}} = \mathbb{E}_{p_{\text{data}}(x,c)}[-\log D_C(c|x)] + \lambda_c \mathbb{E}_{p_g(z,c)}[-d(G(z,c),c)^\beta \log D_C(c|G(z,c))]$$

Improving inference of cGAN (rejection sampling):

- Sample more under-estimated (high GOLD) samples and less over-estimated (low GOLD) samples, to recover the true data distribution
- The acceptance rate can be expressed by the GOLD estimator:[b]

$$r(x) := \frac{1}{M} \exp\left(d(x, c_x)\right) = \frac{1}{M} \exp\left(\log \frac{D_G(x)}{1 - D_G(x)} + \log D_C(c_x|x)\right)$$

Reducing labeling cost of cGAN (active learning):

- From the pool of *unlabeled* samples, choose some samples to ask the *labels*
- It is natural to ask under-estimated (high GOLD) samples, but c_x is unknown!
- Hence, we take an expectation of c_x over the class probability using D_C

$$-\log D_C(c_x|x) \approx \mathbb{E}_{c \sim D_C(c|x)}[-\log D_C(c|x))] = \mathcal{H}[D_C(c|x)]$$

- The GOLD estimator for the unlabeled real samples are given by

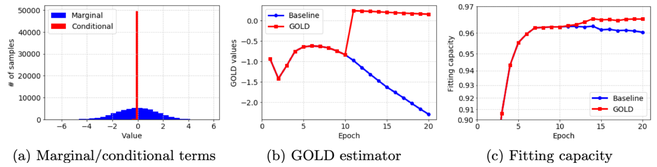
$$d_{\texttt{unlabel}}(x) := \log \frac{D_G(x)}{1 - D_G(x)} + \mathcal{H}[D_C(c|x)]$$

[a]$\beta \geq 0$ is a hyper-parameter to control the level re-weighting and we use $x^\beta = -|x|^\beta$ for $x < 0$.
[b]M is set to be the maximum of $\exp(d_{\texttt{bal}}(x, c_x))$ among samples.

Experiments

We report the fitting capacity[a] [3] as an evaluation metric: it measures the accuracy of *real* samples under a classifier trained with *generated* samples

Figure 2: Training curve of the mean of GOLD estimator and the fitting capacity



(a) Marginal/conditional terms (b) GOLD estimator (c) Fitting capacity

Table 1: Fitting capacity (%) for example re-weighting under various datasets

	MNIST	FMNIST	SVHN	CIFAR-10	STL-10	LSUN
Baseline	96.43±0.17	77.97±1.24	74.43±0.71	36.76±0.99	36.73±0.64	26.35±0.82
GOLD	**96.62**±0.15	**78.34**±1.11	**76.71**±0.94	**37.06**±1.38	**37.65**±0.71	**28.21**±0.86

[a]It is also called as a classification accuracy score (CAS) [4].

Table 2: Fitting capacity (%) for example re-weighting under various levels of supervision

	Dataset	1%	5%	10%	20%	50%	100%
Baseline	SVHN	72.41±1.30	72.99±1.65	73.15±0.96	73.18±1.28	74.04±1.26	74.33±0.71
GOLD	SVHN	**75.01**±1.93	**75.58**±0.86	**75.78**±0.74	**76.04**±1.93	**76.25**±1.40	**76.71**±0.94
Baseline	CIFAR-10	17.99±0.78	18.42±0.71	21.84±1.14	23.13±1.95	35.41±1.03	36.76±0.99
GOLD	CIFAR-10	**18.28**±0.65	**19.15**±0.97	**21.91**±2.56	**23.89**±2.02	34.95±1.11	**37.06**±1.38

Table 3: Fitting capacity (%) for rejection sampling under various datasets

	MNIST	FMNIST	SVHN	CIFAR-10	STL-10	LSUN
Baseline	96.05±0.41	77.94±0.83	73.58±0.72	35.15±0.51	34.33±0.30	26.43±0.14
GOLD	**96.17**±0.63	**78.25**±0.30	**75.06**±0.71	**35.98**±1.15	**35.21**±1.02	**26.79**±0.42

Table 4: Fitting capacity (%) for rejection sampling under CIFAR-10 and various p values

Baseline	p = 0.1	p = 0.3	p = 0.5	p = 0.7	p = 0.9
35.15±0.51	35.80±0.42	35.87±0.61	**35.98**±1.15	35.85±0.53	35.33±0.53

Figure 3: Generated samples with high marginal, conditional, and GOLD values

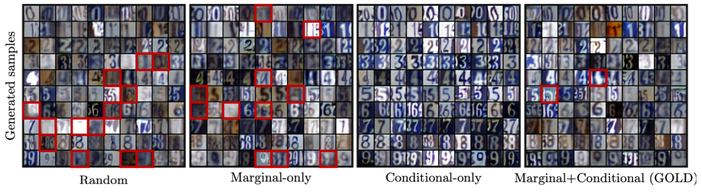
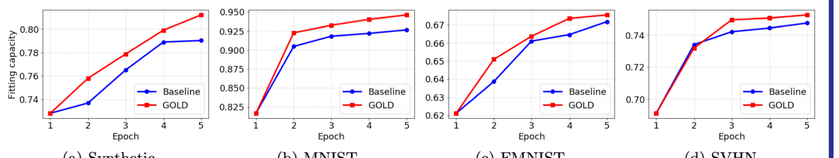


Figure 4: Fitting capacity for active learning under various datasets



(a) Synthetic (b) MNIST (c) FMNIST (d) SVHN